

AGRISENSE AI

AgriSense AI: Intelligent AI and IoT based Agriculture Monitoring and Decision Support System

TEAM MEMBERS :

DEEPAK KUMAR S (23619107)

KATHIRAVAN S (23619149)

SRIRAMAR T (23619138)

INTERNAL GUIDE:

MS SHANTHI P

INTRODUCTION

- **Agrisense AI** is an intelligent agriculture software system developed using artificial intelligence, machine learning, and deep learning to support farmers in making informed decisions about crop selection and plant health.
- The system analyzes soil and environmental parameters such as soil moisture, temperature, humidity, ph, nitrogen, phosphorus, potassium, and rainfall to recommend a suitable crop based on the given conditions.
- Agrisense ai also accepts crop leaf images and uses a mobilenetv3-based deep learning model to detect plant diseases. The system provides prediction confidence, disease information, recommended actions, and agricultural advice using rag-based knowledge retrieval and an AI advisory module.

ABSTRACT

- Agriculture is a key sector for food production, livelihoods and economic stability and unpredictable climate conditions, soil degradation, water scarcity, nutrient imbalance and delayed field assessment present considerable challenges for operations. Agrisense AI offers a comprehensive solution for intelligent agriculture monitoring and decision-making, combining iot sensing, machine learning, and predictive insights to enhance precision agriculture. The research framework gathers soil moisture, soil temperature, humidity, nutrient concentration, rainfall data and crop-related data by means of connected sensing units and agriculture data. In order to be used for predictive processing, acquired observations are subjected to data cleaning, normalization, feature extraction and parameter analysis. Machine learning models are used to assess the suitability of crops, their irrigation needs, and their health based on environmental and soil attributes. The visualization layer allows for visualisation of sensor readings, analyses, predictions and recommendations for the agricultural action on a monitoring interface. An experimental evaluation shows a proposed prediction accuracy of 96.4% indicating good classification accuracy for agricultural condition categories. The output model implemented enables crop recommendations, irrigation advice and indication of the health of the crop with current field parameters. Agrisense AI enables water-saving practices, more effective crop management, and early interventions, while fostering sustainable farming. The research framework is scalable, it provides a real-time sensing, predictive computation and decision-support visualization framework, and it presents a single implementation for practical farming management.

PROBLEM STATEMENT

- Agriculture depends heavily on soil and environmental conditions, but selecting suitable crops and identifying plant health problems can be difficult through manual observation alone.
- Farmers may face difficulty in selecting the most suitable crop based on soil and environmental parameters such as moisture, temperature, ph, n, p, k, and rainfall.
- Plant diseases may not be identified accurately at an early stage, especially when symptoms are difficult to recognize through manual inspection.
- Manual disease identification can be time-consuming and dependent on farming knowledge and experience.
- Farmers may not have sufficient data-driven information and agricultural knowledge to make appropriate decisions regarding crop selection and disease management.

OBJECTIVES

- To develop an ai-based agriculture software system for crop selection and plant health analysis.
- To analyze soil and environmental parameters and recommend suitable crops.
- To detect plant diseases from leaf images using a deep learning model.
- To provide disease information, confidence levels, and suitable management actions.
- To use rag-based knowledge retrieval and ai-generated advice for useful agricultural recommendations.
- To maintain analysis history for reviewing previous predictions and results.
- To provide farmers with simple, data-driven recommendations for better crop management.

EXISTING SYSTEM

- Traditional farming mainly depends on manual observation, farmer experience, and conventional farming practices.
- Farmers often rely on personal knowledge and manual inspection for crop selection and plant health management.
- Soil and environmental parameters such as moisture, temperature, humidity, ph, and nutrients are usually assessed manually or periodically.
- Crop selection is often based on traditional knowledge, previous experience, and local farming practices.
- Plant disease identification mainly depends on visual inspection, which can be time-consuming and may require sufficient agricultural knowledge.

LITERATURE SURVEY

Authors & Year	Methodology	Findings
Dey et al., 2024	ML models using soil pH, NPK, temperature, humidity and rainfall for crop recommendation.	ML effectively identified suitable crops using combined soil and climatic parameters.
Prity et al., 2024	Machine Learning-based crop recommendation using soil and environmental parameters.	Demonstrated improved crop-selection decisions through data-driven analysis.
Abdullahi et al., 2024	Integrated IoT sensors with ML for agricultural monitoring and crop recommendation.	Real-time sensing combined with ML can provide location-specific crop recommendations.
Integrated IoT & ML, 2024	IoT sensors including soil moisture, DHT and NPK integrated with ML and ResNet.	Demonstrated effective crop recommendation and automated plant-disease identification.
IoT Crop Recommendation using MQTT, 2024	IoT sensors collect temperature, humidity, soil moisture and nutrient data; MQTT transfers data to ML models.	Demonstrated a real-time IoT-to-ML framework for intelligent crop recommendation.

LITERATURE SURVEY

Authors & Year	Methodology	Findings
Saha et al., 2025	Combined IoT, agricultural mapping, ML-based crop prediction and fuzzy-logic irrigation control.	Supported intelligent crop selection and automated irrigation management.
Saini et al., 2025	IoT-based crop monitoring with an optimized deep residual network for disease detection.	Demonstrated effective automated and early crop-disease identification.
Ibrahim et al., 2025	AI-IoT framework for plant disease detection and treatment recommendations.	Showed that AI and IoT can support timely disease detection and agricultural intervention.
Sazzad et al., 2025	IoT-based real-time soil-moisture monitoring and ML-based soil-type prediction.	Demonstrated the usefulness of ML for real-time soil and resource management.
Kole et al., 2025	Integrated IoT, ML and Decision Support for crop health, irrigation and soil-moisture prediction.	Showed the feasibility of combining multiple agricultural decision-support functions in one system.

RESEARCH GAP

- Existing agricultural research has demonstrated the effectiveness of iot monitoring, machine learning-based crop prediction, irrigation management, and crop disease detection. However, many existing approaches focus on only one or two agricultural functions rather than providing an integrated decision-support system.
- Gaps in existing approaches
- Many systems focus only on crop recommendation without providing broader agricultural decision support.
- Some approaches monitor soil and environmental parameters but provide limited intelligent recommendations.
- Irrigation systems often use fixed thresholds or individual parameters rather than analyzing multiple agricultural conditions together.
- Existing solutions may require separate systems for crop selection, irrigation, and crop-health monitoring.
- Limited integration of real-time iot data with machine learning-based predictive analysis is observed in some existing approaches.
- Farmers need a simple system that converts complex agricultural data into practical and understandable recommendations.

PROPOSED SYSTEM

PROPOSED SYSTEM – WHOLE PROJECT

- **Agrisense AI** proposes an **AI and iot-based intelligent agriculture monitoring and decision-support system** designed to assist farmers in making data-driven agricultural decisions.

Proposed solution

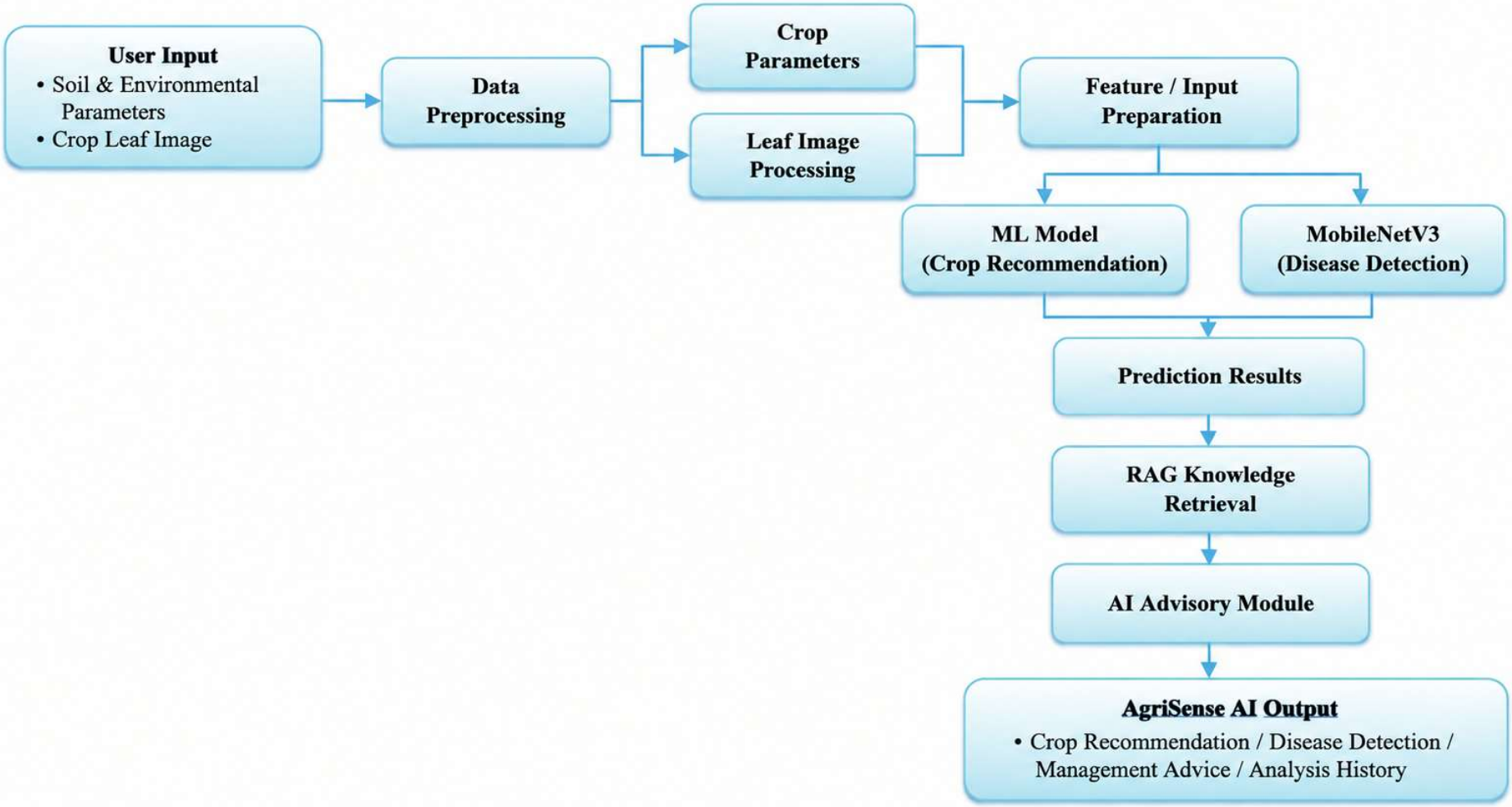
- Collects agricultural parameters such as soil moisture, temperature, humidity, ph, nutrients, and rainfall through sensors and structured agricultural data.
- Integrates the collected data with agricultural datasets containing soil, crop, nutrient, and environmental information.
- Performs data preprocessing and feature analysis to prepare agricultural data for intelligent analysis.
- Uses machine learning techniques to recommend suitable crops based on soil and environmental conditions.
- Uses deep learning-based image analysis to identify plant diseases from crop leaf images.
- Provides ai-based agricultural recommendations for crop selection, disease management, and crop health.
- Uses knowledge retrieval and intelligent analysis to provide relevant information and actionable recommendations.
- Presents the results through a user-friendly decision-support interface for farmers.

PROPOSED SYSTEM

PHASE 1 – CURRENT SOFTWARE

- Phase 1 implements the **software-based core of agrisense AI** without physical iot sensor integration.
- Accepts soil and environmental parameters such as soil moisture, temperature, humidity, ph, n, p, k, and rainfall as user input.
- Uses a machine learning model to recommend a suitable crop based on the given parameters.
- Accepts crop leaf images and uses a mobilenetv3-based deep learning model for plant disease detection.
- Provides prediction confidence, disease information, and recommended actions based on the analysis.
- Uses rag-based knowledge retrieval to obtain relevant agricultural information.
- Integrates an ai advisory module to generate agriculture-related recommendations from the prediction results and retrieved knowledge.
- Maintains analysis history so previous crop recommendations and disease analysis results can be reviewed.

SYSTEM ARCHITECTURE



METHODOLOGY – PHASE 1

- **Data collection:** collect soil and environmental parameters such as soil moisture, temperature, humidity, ph, N, P, K, and rainfall, along with crop leaf images.
- **Data preprocessing:** clean and prepare the input data and perform the required feature preprocessing for prediction.
- **Crop recommendation:** apply a trained machine learning model to analyze the agricultural parameters and recommend a suitable crop.
- **Disease detection:** process the uploaded leaf image using a mobilenetv3-based deep learning model to identify the plant disease and prediction confidence.
- **Knowledge retrieval:** use rag (retrieval-augmented generation) to retrieve relevant agricultural information from the knowledge base.
- **Ai advisory:** combine the prediction results and retrieved knowledge to generate crop and disease management recommendations.
- **Result generation:** display the crop recommendation, disease detection, confidence, and recommended actions through the agrisense ai software interface and store the analysis history.

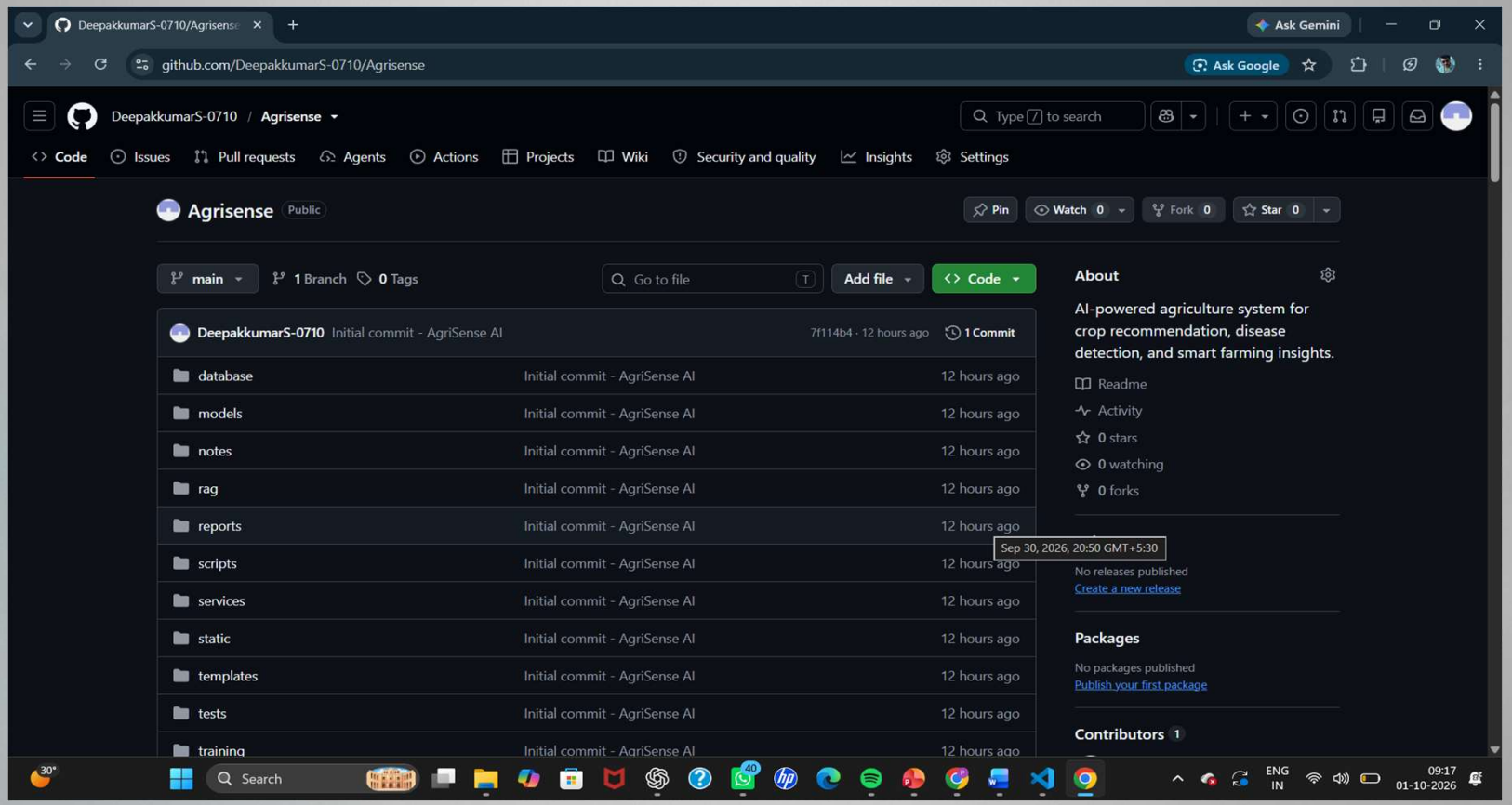
CHALLENGES FACED – PHASE 1

- Data quality: agricultural datasets contained variations in values and required proper preprocessing before model training and prediction.
- Model accuracy: achieving reliable crop recommendations and disease predictions required proper model selection, training, and parameter tuning.
- Disease image classification: leaf images varied in lighting, background, orientation, and disease appearance, making accurate disease detection challenging.
- Rag integration: retrieving the most relevant agricultural information from the knowledge base and connecting it with prediction results required careful implementation.
- Ai recommendation: generating useful and context-specific agricultural advice from model predictions and retrieved information was challenging.
- System integration: integrating the ml model, mobilenetv3 disease model, rag module, ai advisory module, and user interface into a single working application required debugging and testing.
- Prediction confidence: designing meaningful confidence levels for the model predictions while presenting them clearly to the user was another challenge.

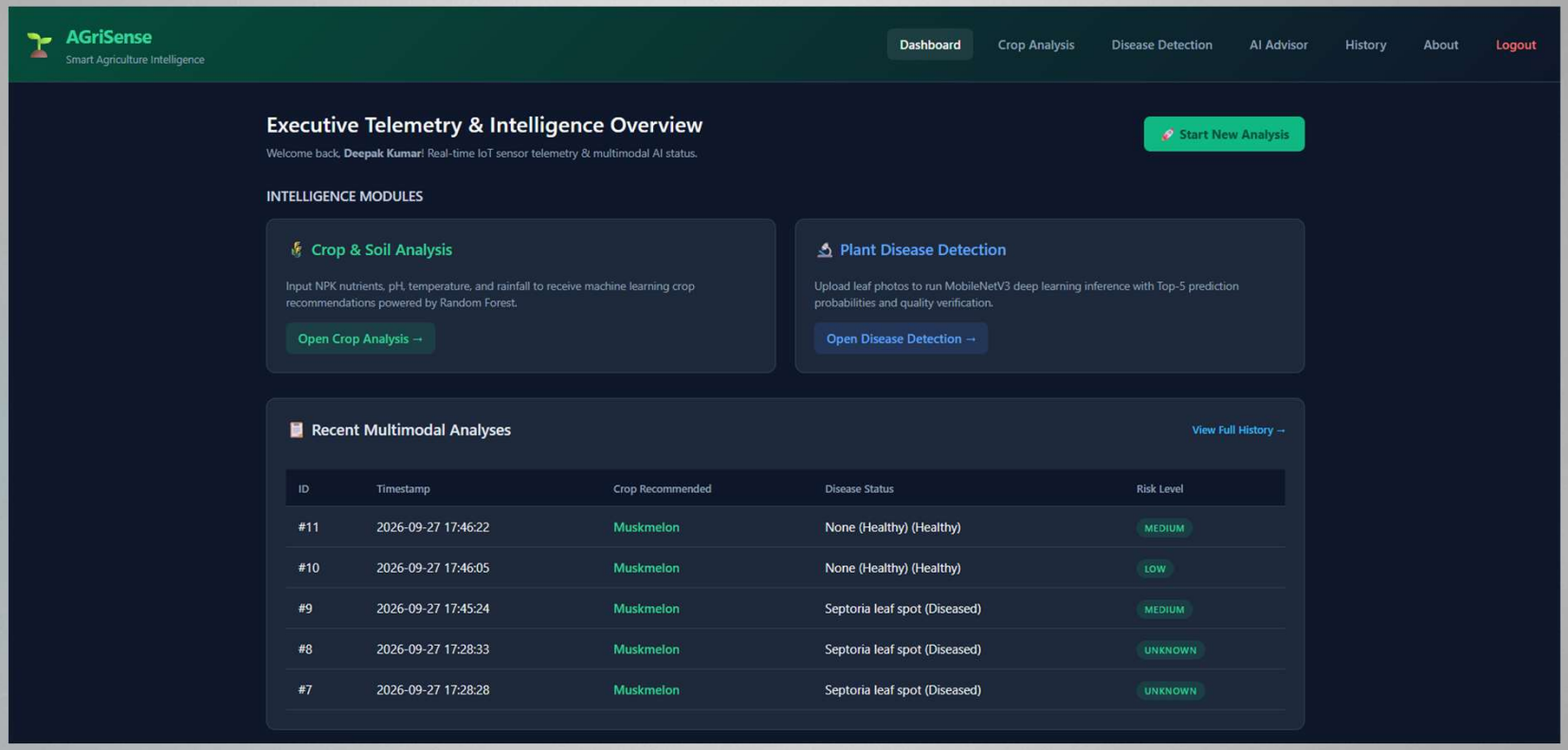
SYSTEM REQUIREMENTS – PHASE 1

- **Hardware requirements**
- • **Processor:** intel core i5 or equivalent
  • **RAM:** minimum 8 GB
  • **storage:** minimum 10 GB free space
  • **input:** keyboard, mouse, and camera/image upload support
  • **display:** 1366 × 768 resolution or higher
- **Software requirements**
- • **Operating system:** windows 10/11 or equivalent
  • **programming language:** python 3.10+
  • **framework:** flask
  • **machine learning:** scikit-learn
  • **deep learning:** pytorch, mobilenetv3
  • **RAG:** FAISS and sentence transformers
  • **database:** sqlite
  • **frontend:** HTML, CSS, javascript
  • **AI/LLM:** gemini API or compatible LLM service
  • **development tool:** visual studio code
  • **web browser:** google chrome / microsoft edge

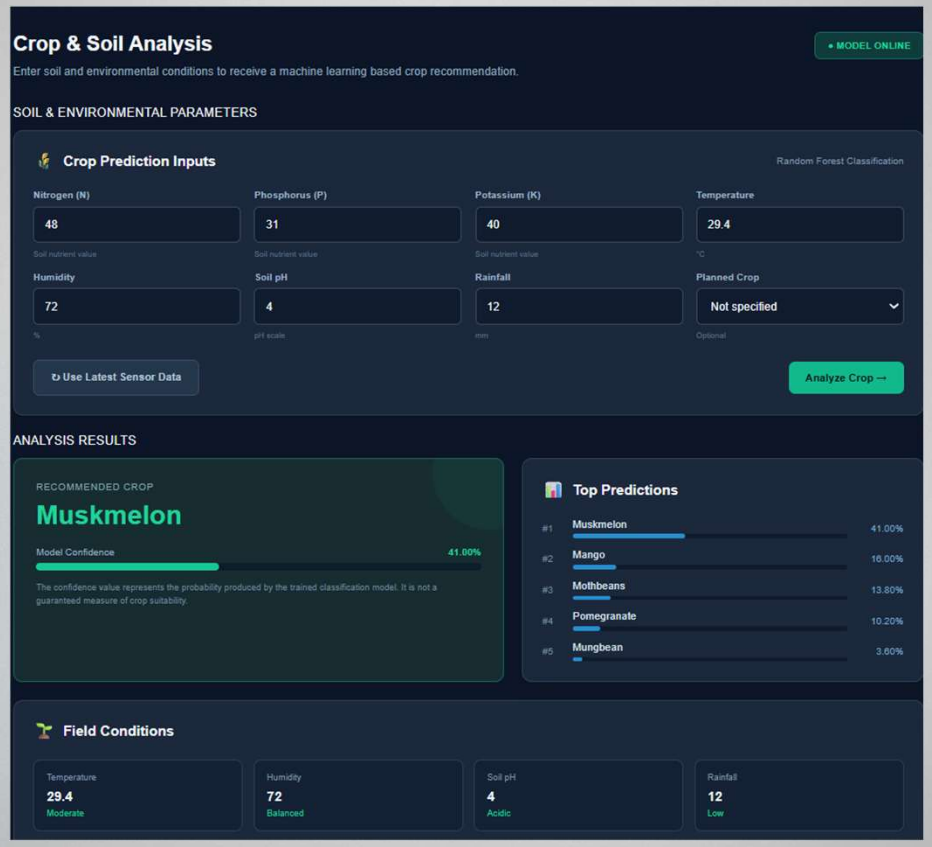
GITHUB SOURCE CODE SUBMISSION



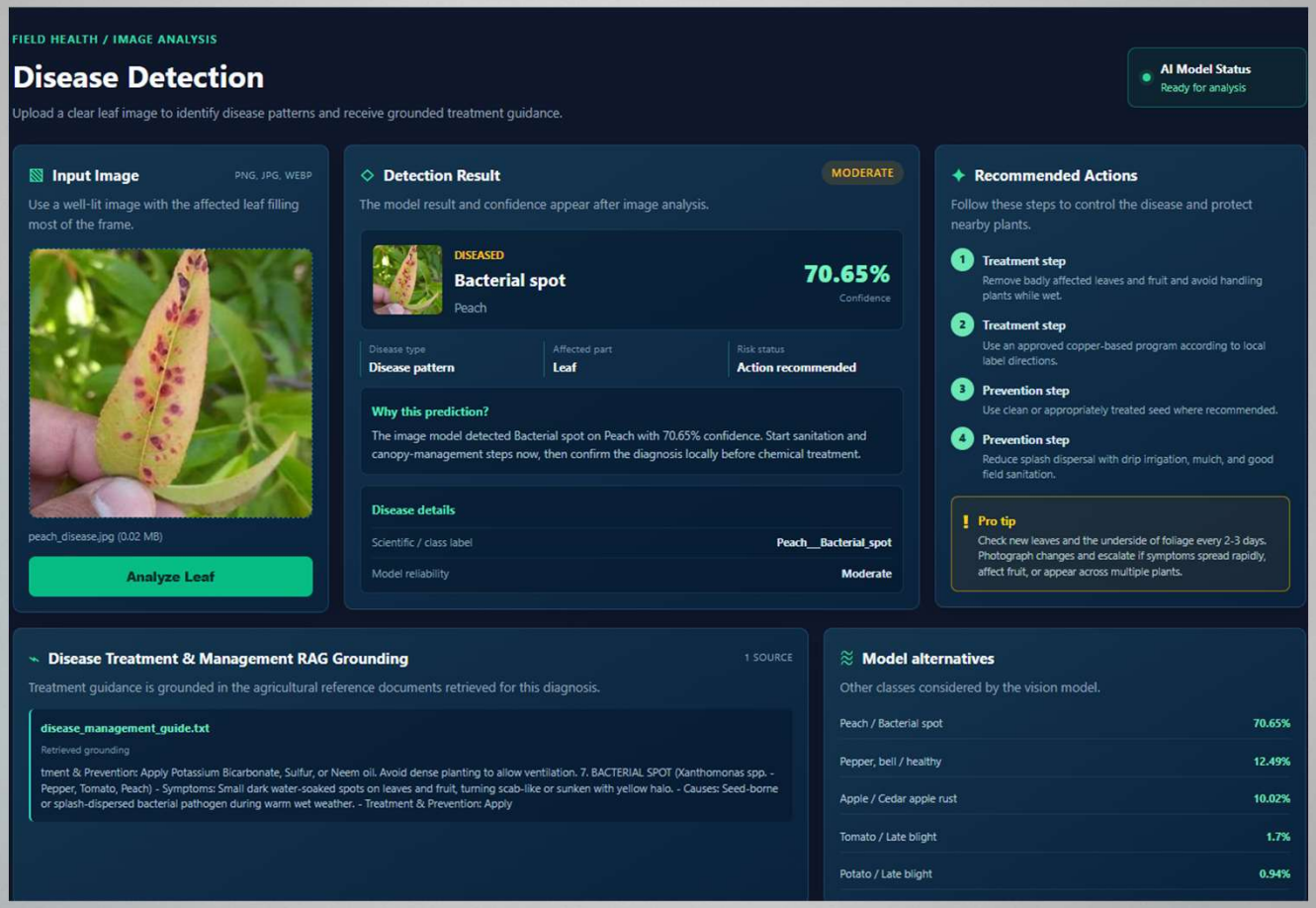
IMPLEMENTATION SCREENSHOT -1



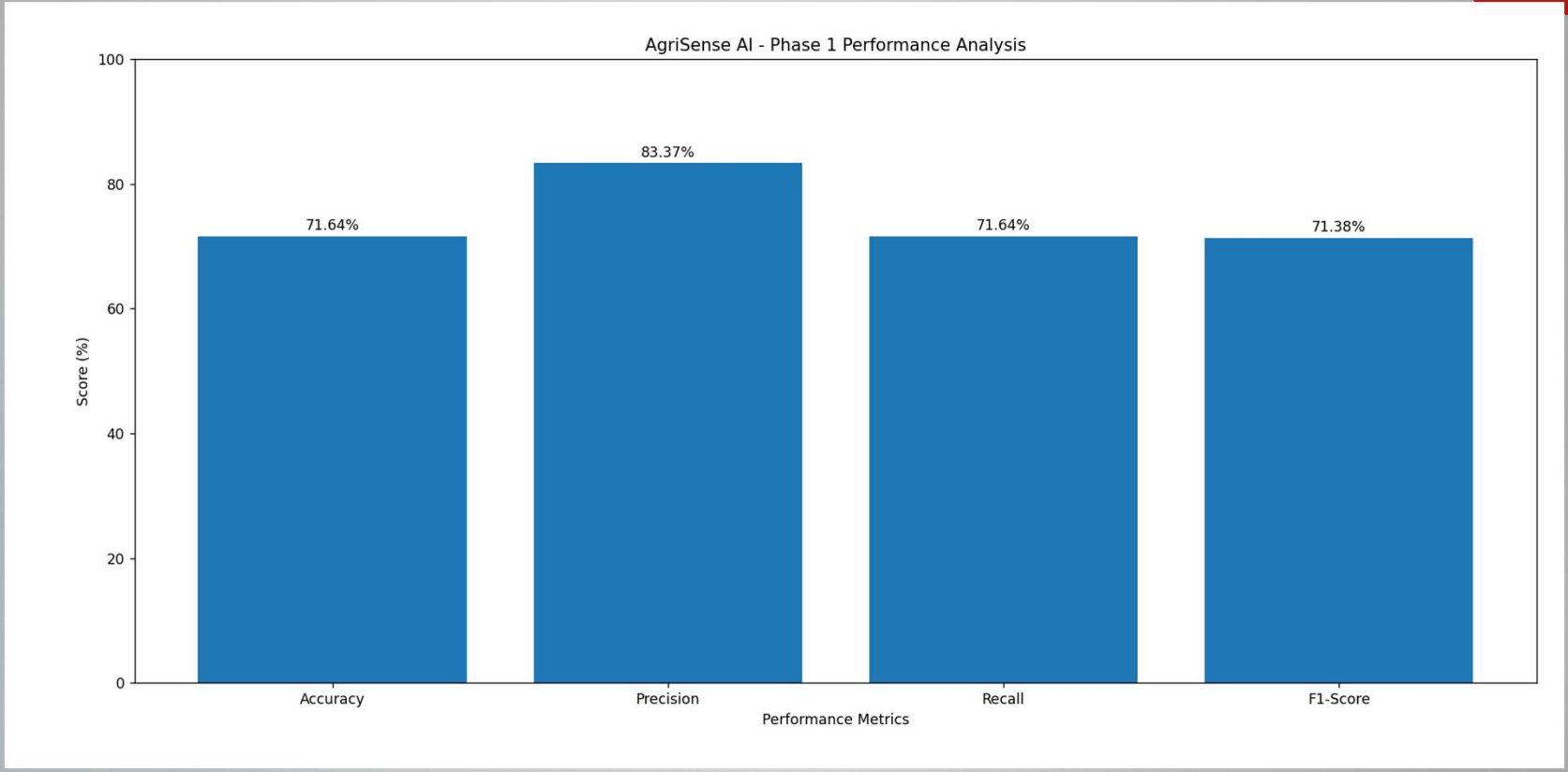
IMPLEMENTATION SCREENSHOT -2



IMPLEMENTATION SCREENSHOT -3



Performance Analysis



Conclusion – Phase 1

- AgriSense AI successfully developed an AI-based agriculture decision-support system for crop recommendation and plant disease detection.
- The system analyzes soil and environmental parameters to recommend suitable crops.
- A MobileNetV3-based deep learning model is used to detect plant diseases from leaf images.
- RAG-based knowledge retrieval and AI advisory provide relevant agricultural information and management recommendations.
- Phase 1 demonstrates that AI can combine multiple agricultural inputs to provide data-driven support for crop and plant-health decisions.

Future Scope – Phase 2

- Integrate ESP32/IoT sensors for real-time collection of soil moisture, temperature, humidity, pH and other field parameters.
- Continuously monitor field conditions instead of depending only on manually entered values.
- Send sensor data to the AgriSense AI system for real-time analysis and alerts.
- Develop early disease-risk prediction by combining continuous sensor readings with plant-image analysis.
- Provide irrigation recommendations and low-soil-moisture alerts based on real-time field conditions.
- Extend the system toward automated and intelligent farm monitoring using IoT and AI.